

One-Sided Passage Beyond the Poisson Boundary: Shell-Averaged Zero Modes and Asymmetric Conductor Rectangles in a Primitive Möbius Tail

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

Fix $h \in \mathbb{Z} \setminus \{0\}$, put $H = \text{rad } |h|$, and let $R = X^{1/2-\delta+o(1)}$ be the finite Ramanujan cutoff in the matched divisor expansion of a primitive fixed-shift Möbius residual. The earlier centered-block argument closes a square $u, v \leq S$ under the condition that the common short prefix satisfies $HS < JX^{-\eta_0}$. We keep that prefix on one row but attach on the other row a new shell $S < v \leq T$, where T may exceed the Poisson length $J = X^{1-\beta+o(1)}$. We write $Q = X^{\beta+o(1)}$, so that $QJ \asymp X$.

The correct object is the exact calibrated-prefix increment. If $P_{m,S}$ is the calibrated residual prefix and $L_{n;S,T} = P_{n,T} - P_{n,S}$, then pointwise

$$P_{m,S}L_{n;S,T} = \sum_{u \leq S} \sum_{S < v \leq T} b_R(u)a(v)\mathbf{1}_{u|mj+h}\mathbf{1}_{v|nj+h} - \delta_H(m)L_{n;S,T},$$

where $a(w) = -\mu(w) \log w$ and $b_R(w) = a(w) - \lambda'_R(w)$, with the inherited coprimality conditions understood. Thus the apparently surviving long single channel cancels algebraically; the remaining scalar drift satisfies $\delta_H(\ell d_0) \ll X^{o(1)}/L$. The joint nonzero channel admits an asymmetric exact-conductor rectangle. If $S = X^{s+o(1)}$, $T = X^{t+o(1)}$, and $s \leq t$, its certified saving is

$$\eta_{\text{asym}}(\beta; s, t) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 1 - 2s - 2t}{2}, \frac{3 - \beta - s - 5t}{4} \right\}.$$

The additive zero frequency is not covered by the nonzero Ramanujan–Gauss estimate. We prove a separate shell theorem. For each off-diagonal row pair, split the shared factor $g = (u, v) \mid m - n$ at $SX^{-\kappa}$, where $0 < \kappa < s$. For small g , uniform Möbius calibration at the two endpoints S/g and T/g cancels the common main constant. For large g , divisibility of the nonzero row determinant gives a factor X^κ/S . If $\mathcal{Z}_1$ is the absolute row-pair mass, then for every fixed $A > 0$ the resulting zero-mode bound is

$$\ll J\mathcal{Z}_1 \left\{ (\log X)^{-A} + X^\kappa S^{-1} X^{o(1)} \right\}.$$

This mass factor is not hidden: the usual soft estimate $\mathcal{Z}_1 \ll Q^2 X^\varepsilon$ does not turn the first term into logarithmic saving. When $\eta_{\text{asym}} > 0$ with fixed margins and the stated interfaces hold, the explicit fixed-projective-mass hypothesis $\mathcal{Z}_1 \ll Q^2 (\log X)^K$ makes the entire calibrated mixed increment $O_B(XQ(\log X)^{-B})$ for every fixed $B > 0$. With only the soft mass hypothesis, test functions with $\widehat{F}(0) = 0$ give fixed-power closure; the displayed zero mode remains the sole mass-sensitive term in general.

For $S = R$ and $T = RX^\xi$, the joint saving becomes

$$\min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 4\delta - 1}{2} - \xi, \frac{6\delta - \beta - 5\xi}{4} \right\}.$$

Consequently there is a strict one-sided passage beyond $T = J$ when

$$\frac{1}{2} + \delta - \beta < \xi < \min \left\{ \frac{\beta + 4\delta - 1}{2}, \frac{6\delta - \beta}{5} \right\},$$

provided $\beta < 2/3$, $\beta < 1/2 + \delta$, and all inequalities have fixed margins. This is an incremental mixed-shell estimate, not a closure of the full residual, a Hardy–Littlewood asymptotic, a twin-prime lower bound, or a general breach of sieve parity.

Keywords: primitive Möbius tail; Poisson boundary; mixed divisor shell; shell-averaged zero mode; exact conductors; tensor large sieve; sieve parity.

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1 Introduction

The primitive residual program has reached two different analytic boundaries. The first is a conductor boundary: after exact shared-factor rectangularization, two complementary estimates for the same conductor cell can produce a fixed power saving. The second is a Poisson boundary: the four centered divisibility channels have periods $[u, v]$, u , v , and 1, so a single channel is completed by smoothness only when its period is shorter than the orbit length.

TPC-20 derived the exact four-channel formula and completed both single channels under

$$H \max(u, v) \leq JX^{-\eta_0}. \quad (1)$$

TPC-22 closed the resulting joint short–short packet by exact conductors, and TPC-23 moved the common cutoff from $R = X^{1/2-\delta+o(1)}$ to $S > R$, still subject to (1) [4–6]. The longer single channel, not the conductor tensor, stops that centered-block argument at $S < J$.

This paper moves only one divisor leg. Let

$$R \leq S < T \leq U, \quad S = X^{s+o(1)}, \quad T = X^{t+o(1)}, \quad 0 < s \leq t, \quad (2)$$

where $U = X^{O(1)}$ is the finite target horizon. Put

$$a(v) = -\mu(v) \log v, \quad b_R(u) = a(u) - \lambda'_R(u),$$

with $\lambda'_R(u) = 0$ for $u > R$. The new divisor region is the incremental rectangle

$$u \leq S, \quad S < v \leq T. \quad (3)$$

It is essential that the second leg is a complete Möbius shell, not an isolated long modulus or an arbitrary dyadic subset.

1.1 The calibrated increment is the stable object

Let $P_{m,W}(j)$ be the locally calibrated residual prefix defined in [equation \(45\)](#). Its exact finite identity is

$$P_{m,W}(j) = \sum_{\substack{w \leq W \\ w|mj+h}} b_R(w) - \delta_H(m), \quad \delta_H(\ell d_0) \ll_{\epsilon,h} \frac{X^\epsilon}{L}. \quad (4)$$

Because the finite model ends at R , the increment is

$$L_{m;S,T}(j) := P_{m,T}(j) - P_{m,S}(j) = \sum_{\substack{S < w \leq T \\ w|mj+h}} a(w). \quad (5)$$

We study the separated packet

$$\mathcal{I}_{S,T} := \sum_{m,n} x_m y_n \mathbf{m}(m,n) \sum_{\substack{j \in \mathbb{Z} \\ (j,H)=1}} F(j/J) P_{m,S}(j) L_{n,S,T}(j). \quad (6)$$

Its transpose is identical after exchanging the row labels.

The pointwise product in (6) has the exact expansion

$$P_{m,S}(j) L_{n,S,T}(j) = \sum_{\substack{u \leq S \\ u|mj+h}} b_R(u) \sum_{\substack{S < v \leq T \\ v|nj+h}} a(v) - \delta_H(m) L_{n,S,T}(j). \quad (7)$$

This identity is the first structural result. If one starts instead from two bare centered divisor sums, a long v -single Poisson channel survives and is multiplied by a merely logarithmically calibrated short density. In the genuine prefix increment, the scalar calibration terms restore the omitted channels and cancel that long single exactly. What remains is one joint congruence lattice and the already power-saving row drift in (4).

1.2 The asymmetric conductor rectangle

For one compatible pair, write

$$g = (u, v), \quad u = gc, \quad v = gd, \quad q = [u, v] = gcd.$$

The TPC-22 character coordinates and weighted exact-conductor large sieve extend to the rectangle $u \leq S$, $v \leq T$. If $Q = X^{\beta+o(1)}$, $J = X^{1-\beta+o(1)}$, then every joint nonzero conductor cell has one of two complementary bounds. Their three-cell ledger gives

$$\eta_{\text{asym}}(\beta; s, t) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 1 - 2s - 2t}{2}, \frac{3 - \beta - s - 5t}{4} \right\}. \quad (8)$$

Thus the joint nonzero discrepancy has a fixed power saving whenever $\eta_{\text{asym}} > 0$ with a fixed margin. No hypothesis $T < J$ or $L > 2T$ occurs in this conductor estimate.

1.3 The physical zero mode and the mass ledger

The induced-Gauss and Ramanujan-frequency estimates explicitly exclude additive frequency zero. For an off-diagonal row pair put $\Delta = m - n$. The zero-frequency kernel of the joint term in (7) is

$$K_{S,T}^0(m, n) = \sum_{\substack{u \leq S, S < v \leq T \\ (u,mH)=(v,nH)=1}} \frac{b_R(u)a(v)}{uv} (u, v) \mathbf{1}_{(u,v)|\Delta}. \quad (9)$$

Choose $0 < \kappa < s$. Write $u = gc$, $v = gd$, with $g \mid \Delta$. If $g \leq SX^{-\kappa}$, then $S/g \geq X^\kappa$. For fixed g, c , the inner d -sum is a difference of uniformly calibrated Möbius sums at T/g and S/g ; their common main constant cancels. If $g > SX^{-\kappa}$, then the off-diagonal mask gives

$$\sum_{\substack{g \mid \Delta \\ g > SX^{-\kappa}}} \frac{1}{g} \leq \frac{X^\kappa}{S} \tau(|\Delta|). \quad (10)$$

Consequently, for every fixed $A > 0$,

$$K_{S,T}^0(m, n) \ll_{A,\kappa} (\log X)^{-A} + X^{-s+\kappa+o(1)}. \quad (11)$$

Let

$$\mathcal{Z}_1 = \sum_{m,n} |x_m y_n \mathbf{m}(m, n)|. \quad (12)$$

The full physical zero mode is therefore bounded by

$$J \mathcal{Z}_1 \left\{ (\log X)^{-A} + X^{-s+\kappa+o(1)} \right\}. \quad (13)$$

The factor $\mathcal{Z}_1$ is deliberate. From the soft estimate $\mathcal{Z}_1 \ll Q^2 X^\epsilon$, one may close the large- g term by choosing $\epsilon < s - \kappa$, but one may not convert $X^\epsilon (\log X)^{-A}$ into logarithmic saving. Under the stronger and independently checkable projective hypothesis

$$\mathcal{Z}_1 \ll Q^2 (\log X)^K \quad (14)$$

for a fixed K , the exponent A can be chosen after B, K , and (13) is $O_B(XQ(\log X)^{-B})$. This distinction is part of the theorem, not an X^ϵ -notation convention. If $\hat{F}(0) = 0$, the entire mass-sensitive term vanishes and the soft row estimates suffice.

1.4 A genuine passage beyond J

Specialize to

$$S = R = X^{1/2-\delta+o(1)}, \quad T = RX^\xi. \quad (15)$$

Then (8) becomes

$$\eta_{\text{asym}}(\beta, \delta, \xi) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 4\delta - 1}{2} - \xi, \frac{6\delta - \beta - 5\xi}{4} \right\}. \quad (16)$$

The old Poisson face is

$$\xi_P = \frac{1}{2} + \delta - \beta, \quad T = J \iff \xi = \xi_P.$$

Suppose

$$\beta < \frac{2}{3}, \quad \beta < \frac{1}{2} + \delta, \quad (17)$$

with fixed margins. Then the strict interval

$$\boxed{\frac{1}{2} + \delta - \beta < \xi < \min \left\{ \frac{\beta + 4\delta - 1}{2}, \frac{6\delta - \beta}{5} \right\}} \quad (18)$$

is a certified one-sided passage beyond J . A convenient nonempty region is

$$\delta > \frac{1}{10}, \quad \frac{5}{8} - \frac{\delta}{4} < \beta < \min \left\{ \frac{2}{3}, \frac{1}{2} + \delta \right\}. \quad (19)$$

For example, $(\delta, \beta, \xi) = (0.15, 0.62, 0.04)$ gives

$$R = X^{0.35}, \quad J = X^{0.38}, \quad T = X^{0.39}, \quad \eta_{\text{asym}} = 0.02.$$

1.5 Scope and organization

The theorem treats the incremental mixed shell $[1, S] \times (S, T]$, and by transposition its mirror. Under (14) it closes that calibrated increment, including physical frequency zero. Under only the generic soft mass it proves the fixed-power nonzero theorem and the exact mass-sensitive zero estimate (13); this is not silently promoted to full closure. The result does not re-prove the old base square, control the both-ultra region, restore every endpoint of the TPC-18 residual [2], or produce a positive prime-pair main term.

Section 2 fixes the exact prefix and row interfaces. Section 3 explains the four-channel cancellation behind (7). Section 4 proves (11). Section 5 extends the conductor tensor to a rectangle, and section 6 proves the ledger and passage window. Exact splicing, the failed pointwise shortcut, and the scope ledger are in section 7. The certificate and conclusion are in section 8.

2 Primitive rows, calibrated prefixes, and the mixed packet

Fix $h \in \mathbb{Z} \setminus \{0\}$, put $H = \text{rad } |h|$, and choose

$$R = \lfloor X^{1/2-\delta} \rfloor, \quad 0 < \delta < \frac{1}{2}, \quad R \geq H^2. \quad (20)$$

In one separated primitive row packet,

$$Q = LD, \quad QJ \asymp X, \quad L > \max\{2R, 2|h|\}, \quad (21)$$

and

$$m = \ell d_0 \asymp Q, \quad \ell \in [L/2, 2L] \text{ prime}, \quad d_0 \in [D, 2D], \quad d_0 \leq R^{1/2}. \quad (22)$$

The inherited primitive conditions include $(d_0, H) = 1$, $(j, H) = 1$, and $d_0 H \leq R$. Hence

$$(w, mH) = 1 \quad \text{for every } w \mid mj + h. \quad (23)$$

The separated row weights satisfy

$$\|x\|_\infty + \|y\|_\infty \ll_\varepsilon X^\varepsilon, \quad \sum_m |x_m|^2 + \sum_n |y_n|^2 \ll QX^\varepsilon, \quad (24)$$

and their absolute pair mass is denoted by

$$\mathcal{Z}_1 := \sum_{m,n} |x_m y_n \mathbf{m}(m, n)| \ll_\varepsilon Q^2 X^\varepsilon. \quad (25)$$

Here $\mathbf{m}$ is the actual bounded generic row-pair mask. We use only

$$|\mathbf{m}(m, n)| \leq 1, \quad \mathbf{m}(m, m) = 0, \quad \mathbf{m} \text{ is independent of all divisor variables.} \quad (26)$$

The last two properties are essential in different places: the conductor tensor uses divisor independence, while the large- g zero-mode argument uses $m - n \neq 0$.

For the strongest logarithmic conclusion we state separately the fixed-projective-mass hypothesis

$$\mathcal{Z}_1 \ll Q^2 (\log X)^K \quad (27)$$

for some fixed $K \geq 0$. If a scalar packet is reassembled from a projective sum $\sum_\nu x^{(\nu)} \otimes y^{(\nu)}$, the left side of (27) is replaced by $\sum_\nu \|x^{(\nu)}\|_1 \|y^{(\nu)}\|_1$. The soft estimate (25) does not imply (27); the two hypotheses are kept distinct throughout.

2.1 Fourier and fiber notation

We use

$$e(t) = e^{2\pi i t}, \quad e_q(t) = e(t/q), \quad c_q(r) = \sum_{\xi \in (\mathbb{Z}/q\mathbb{Z})^\times} e_q(r\xi), \quad (28)$$

and

$$\widehat{F}(t) = \int_{\mathbb{R}} F(x) e(-tx) dx. \quad (29)$$

Write $\mathcal{U}_q = (\mathbb{Z}/q\mathbb{Z})^\times$ and let $\mathfrak{X}(q)$ be its group of Dirichlet characters. Whenever $(H, q) = 1$, the symbol $\overline{H}$ denotes the inverse of H modulo q . Finally, the centered divisor indicator is

$$C_{m,w}(j) = \mathbf{1}_{w \mid mj+h} - \frac{1}{w}. \quad (30)$$

2.2 Divisor coefficients and exact densities

Put

$$a(w) = -\mu(w) \log w. \quad (31)$$

The finite Ramanujan model has the divisor expansion

$$\Lambda_R(n) = \sum_{w|n} \lambda'_R(w), \quad \lambda'_R(w) = w \sum_{b \leq R/w} \frac{\mu(wb)\mu(b)}{\varphi(wb)} \quad (w \leq R), \quad (32)$$

and $\lambda'_R(w) = 0$ for $w > R$. Define

$$b_R(w) = a(w) - \lambda'_R(w). \quad (33)$$

All three coefficients are supported on squarefree integers, and uniformly for $w \leq X^{O(1)}$,

$$|a(w)| + |b_R(w)| + |\lambda'_R(w)| \ll_\varepsilon X^\varepsilon. \quad (34)$$

For a row $m = \ell d_0$, the exact finite-model and calibrated prime densities are

$$\rho_{M,H,R}(m) = \frac{H}{\varphi(H)} \frac{d_0}{\varphi(d_0)}, \quad \rho_{P,H}(m) = \frac{mH}{\varphi(mH)}. \quad (35)$$

TPC-19–20 prove the finite identity

$$\sum_{\substack{w \leq R \\ (w, mH)=1}} \frac{\lambda'_R(w)}{w} = \rho_{M,H,R}(m) \quad (36)$$

and the row drift formula

$$\delta_H(m) := \rho_{P,H}(m) - \rho_{M,H,R}(m) = \frac{H}{\varphi(H)} \frac{d_0}{\varphi(d_0)(\ell-1)} \ll_{\varepsilon,h} \frac{X^\varepsilon}{L}. \quad (37)$$

These are exact finite algebra, not prime distribution in a growing progression [3, 4].

2.3 Uniform endpoint calibration

For $M \geq 1$ and $Y \geq 1$, set

$$A_M(Y) = \sum_{\substack{w \leq Y \\ (w, M)=1}} \frac{-\mu(w) \log w}{w}, \quad B_M(Y) = \sum_{\substack{w \leq Y \\ (w, M)=1}} \frac{\mu(w)}{w}. \quad (38)$$

Lemma 2.1 (Uniform calibration and its companion). *For every fixed $A, C > 0$, uniformly for $Y \geq 3$ and $M \leq Y^C$,*

$$A_M(Y) = \frac{M}{\varphi(M)} + O_{A,C} \left(\frac{M}{\varphi(M)} (\log Y)^{-A} \right), \quad (39)$$

$$B_M(Y) \ll_{A,C} \frac{M}{\varphi(M)} (\log Y)^{-A}. \quad (40)$$

Consequently, if $Y_2 \geq Y_1 \geq X^\kappa$ and $M \leq X^{C_0}$, where $C_0, \kappa > 0$ are fixed, then for every $A > 0$,

$$A_M(Y_2) - A_M(Y_1) \ll_{A,C_0,\kappa} (\log X)^{-A+1}, \quad (41)$$

$$B_M(Y_2) - B_M(Y_1) \ll_{A,C_0,\kappa} (\log X)^{-A+1}, \quad (42)$$

uniformly with constants depending on A, C_0, κ .

Proof. Equation (39) is the uniform Möbius calibration theorem of TPC-20 [4]. We recall why the same proof gives (40). Let

$$\mathcal{S}(M) = \{k \geq 1 : p \mid k \implies p \mid M\}, \quad G_M(s) = \prod_{p \mid M} (1 - p^{-s})^{-1}.$$

The exact convolution

$$\mu(n) \mathbf{1}_{(n,M)=1} = (\mu * \mathbf{1}_{\mathcal{S}(M)})(n)$$

gives

$$B_M(Y) = \sum_{\substack{k \leq Y \\ k \in \mathcal{S}(M)}} \frac{1}{k} \sum_{r \leq Y/k} \frac{\mu(r)}{r}. \quad (43)$$

The classical zero-free region for $\zeta(s)$ implies

$$\sum_{r \leq Z} \frac{\mu(r)}{r} \ll_A (\log Z)^{-A} \quad (Z \geq 3).$$

Split (43) at $k = Y/\exp(\sqrt{\log Y})$. The central range is bounded with the displayed zero-free-region estimate. The remaining range is handled by the same Rankin bound for $G_M(1 - \eta)$ used in the proof of (39); it is uniform for $M \leq Y^C$. This proves (40).

Apply the first part with $C = C_0/\kappa$, and subtract (39) at Y_2 and Y_1 : the common main term $M/\varphi(M)$ cancels. Do the same with (40), use $M/\varphi(M) \ll \log \log(3M)$, and increase A to absorb this factor. Since $\log Y_1 \geq \kappa \log X$, the interval estimates follow. The only external analytic input is the classical zero-free region; see Iwaniec and Kowalski [1, Chapter 5]. $\square$

2.4 Complete prefixes

For $R \leq W \leq U$, define

$$\vartheta_W(w) = b_R(w) \mathbf{1}_{w \leq W}, \quad E_W(m) = A_{mH}(W) - \rho_{P,H}(m), \quad (44)$$

and

$$P_{m,W}(j) = E_W(m) + \sum_{\substack{w \leq W \\ (w,mH)=1}} b_R(w) C_{m,w}(j). \quad (45)$$

Proposition 2.2 (Exact calibrated-prefix identities). *For every $R \leq W \leq U$,*

$$\sum_{\substack{w \leq W \\ (w,mH)=1}} \frac{b_R(w)}{w} = \delta_H(m) + E_W(m), \quad (46)$$

$$P_{m,W}(j) = \sum_{\substack{w \leq W \\ w \mid mj+h}} b_R(w) - \delta_H(m). \quad (47)$$

$$P_{m,T}(j) - P_{m,S}(j) = \sum_{\substack{S < w \leq T \\ w \mid mj+h}} a(w). \quad (48)$$

If U contains every positive target in the packet, then

$$P_{m,U}(j) = \Lambda(mj+h) - \Lambda_R(mj+h) - \delta_H(m), \quad (49)$$

which is the locally centered residual signal of TPC-20.

Proof. Since λ'_R is supported on $w \leq R$,

$$\sum_{\substack{w \leq W \\ (w, mH)=1}} \frac{b_R(w)}{w} = A_{mH}(W) - \rho_{M, H, R}(m).$$

Insert (35) to obtain (46). Expanding $C_{m, w} = \mathbf{1}_{w|mj+h} - 1/w$ in (45) then proves (47). Subtract the formulas at T and S ; on $w > R$, $b_R(w) = a(w)$, proving (48). At the target horizon, the von Mangoldt divisor identity and (32) give (49). $\square$

Lemma 2.3 (Prefix coefficient mass). *For every fixed $C > 0$, uniformly for $R \leq W \leq X^C$,*

$$\sum_{w \leq W} \frac{|b_R(w)|}{w} \ll_C (\log(2X))^3. \quad (50)$$

Moreover, uniformly in squarefree $g \leq W$,

$$\sum_{\substack{c \leq W/g \\ (c, g)=1}} \frac{|b_R(gc)|}{c} \ll_C (\log(2X))^4. \quad (51)$$

Proof. The a -part is bounded by a harmonic logarithmic sum. For the finite-model part use the positive formula

$$|\lambda'_R(w)| \leq \frac{w}{\varphi(w)} \sum_{b \leq R/w} \frac{\mu(b)^2}{\varphi(b)} \quad (w \text{ squarefree})$$

and sum the resulting nonnegative double divisor series. This gives $\sum_{w \leq R} |\lambda'_R(w)|/w \ll (\log(2R))^2$, hence (50). In (51), use $\varphi(gc) = \varphi(g)\varphi(c)$ on $(g, c) = 1$, and bound the two remaining harmonic divisor sums. The harmless factor $g/\varphi(g)$ is absorbed by one additional logarithm. $\square$

2.5 The calibrated mixed increment and inherited interfaces

Let F range over a uniformly bounded subset of $C_c^\infty(\mathbb{R})$, and put

$$L_{m; S, T}(j) := P_{m, T}(j) - P_{m, S}(j) = \sum_{\substack{S < w \leq T \\ w|mj+h}} a(w). \quad (52)$$

The main packet is

$$\mathcal{I}_{S, T} := \sum_{m, n} x_m y_n \mathbf{m}(m, n) \sum_{\substack{j \in \mathbb{Z} \\ (j, H)=1}} F(j/J) P_{m, S}(j) L_{n; S, T}(j). \quad (53)$$

Its natural scale is

$$Q^2 J \asymp XQ. \quad (54)$$

For a compatible pair u, v , let $q = [u, v]$ and let $\rho_{m, n}^{u, v} \in \mathcal{U}_q$ be the joint residue class determined by

$$u \mid m\rho_{m, n}^{u, v} + h, \quad v \mid n\rho_{m, n}^{u, v} + h.$$

Define the exact joint fiber

$$Z_{q; S, T}(\xi) := \sum_{\substack{u \leq S, \ S < v \leq T \\ [u, v]=q}} b_R(u) a(v) \sum_{\substack{m, n \\ (u, v) | m-n \\ \rho_{m, n}^{u, v} \equiv \xi \pmod{q}}} x_m y_n \mathbf{m}(m, n), \quad (55)$$

with all row and target-primitivity restrictions retained. Put

$$\overline{Z}_{q;S,T} = \frac{1}{\varphi(q)} \sum_{\xi \in \mathcal{U}_q} Z_{q;S,T}(\xi), \quad D_{q;S,T} = Z_{q;S,T} - \overline{Z}_{q;S,T}. \quad (56)$$

The joint nonzero scalar packet is

$$\mathcal{J}_{q;S,T}^{\neq 0} = \frac{J}{Hq} \sum_{r \neq 0} c_H(r) \widehat{F}\left(\frac{rJ}{Hq}\right) \sum_{\xi \in \mathcal{U}_q} Z_{q;S,T}(\xi) e_q(r\overline{H}\xi). \quad (57)$$

Its mean and discrepancy parts are obtained by replacing $Z_{q;S,T}$ with $\overline{Z}_{q;S,T}$ and $D_{q;S,T}$, respectively.

We assume explicitly the TPC-20 no-wrap condition

$$J \geq 4HX^{\epsilon_0} \quad (58)$$

for some fixed $\epsilon_0 > 0$ [4]. Fix once and for all

$$0 < \omega < \epsilon_0. \quad (59)$$

To state the Fourier-tail interface quantitatively, define

$$\mathcal{T}_\omega := \sum_{q \leq ST} \frac{J}{Hq} \sum_{|r| > X^\omega Hq/J} |c_H(r)| \left| \widehat{F}\left(\frac{rJ}{Hq}\right) \right| \sum_{\xi \in \mathcal{U}_q} |Z_{q;S,T}(\xi)|. \quad (60)$$

The inherited polynomial tail-mass hypothesis is that for every fixed $A > 0$, the Fourier decay order may be chosen so that

$$\mathcal{T}_\omega \ll_{A,F,H} X^{-A} \mathcal{Z}_1. \quad (61)$$

This is the precise tail discarded in the cell reassembly below. The fixed inequality $\omega < \epsilon_0$, not an A -dependent window, is what preserves no-wrap on the retained frequencies. Moreover,

$$\sum_{\xi \in \mathcal{U}_q} |D_{q;S,T}(\xi)| \leq 2 \sum_{\xi \in \mathcal{U}_q} |Z_{q;S,T}(\xi)|,$$

so (61) controls the discrepancy tail as well as the undecomposed joint tail. On the retained window, $|r| \leq X^\omega Hq/J < q/4$ for all sufficiently large X , by (58) and (59).

For the four-channel interpretation in section 3 we also record the short-side margin

$$HS \leq JX^{-\eta_0} \quad (62)$$

for some fixed $\eta_0 > 0$. The exact calibrated recombination and the conductor theorem themselves do not require (62). No comparison between T and J , or between L and T , is imposed.

Proposition 2.4 (Exact joint-minus-drift recombination). *Pointwise in the two rows and the orbit variable,*

$$P_{m,S}(j)L_{n;S,T}(j) = \sum_{\substack{u \leq S \\ u|mj+h}} b_R(u) \sum_{\substack{S < v \leq T \\ v|nj+h}} a(v) - \delta_H(m)L_{n;S,T}(j). \quad (63)$$

Moreover, for every fixed $\epsilon > 0$, the drift contribution to (53) is

$$\ll_{\epsilon,F,h} J \mathcal{Z}_1 L^{-1} X^\epsilon. \quad (64)$$

Proof. The identity follows at once from (47) and (52). Also

$$|L_{n;S,T}(j)| \leq \sum_{v|nj+h} |a(v)| \leq \tau(nj+h) \log(2X) \ll_{\epsilon,F} X^\epsilon.$$

There are $O_F(J)$ orbit points, and (37) and the definition of $\mathcal{Z}_1$ prove (64) after renaming ϵ . $\square$

3 Four channels and exact calibrated recombination

This section explains why a bare centered mixed rectangle has an apparently dangerous long single channel, and why that channel is absent from the exact calibrated increment. Fix squarefree $u \leq S < v \leq T$, assume

$$(u, mH) = (v, nH) = 1, \quad (65)$$

and put

$$g = (u, v), \quad u = gc, \quad v = gd, \quad q = [u, v] = gcd. \quad (66)$$

Then g, c, d are pairwise coprime. Write

$$\iota_{m,n}(g) = \mathbf{1}_{g|m-n}. \quad (67)$$

When $\iota_{m,n}(g) = 1$, the two divisibility congruences determine a unit class $\rho \bmod q$. Here $\overline{m}$ and $\overline{n}$ denote the inverses of m modulo u and of n modulo v , respectively. The two individual congruences determine

$$\rho_1 \equiv -h\overline{m} \pmod{u}, \quad \rho_2 \equiv -h\overline{n} \pmod{v}. \quad (68)$$

3.1 The two surviving channels of a bare centered pair

Theorem 3.1 (Two surviving channels). *Assume (62). Uniformly for the pairs above and every prescribed $A > 0$,*

$$\begin{aligned} & \sum_{\substack{j \in \mathbb{Z} \\ (j,H)=1}} F(j/J) C_{m,u}(j) C_{n,v}(j) \\ &= \frac{J}{Huv} \left[g\iota_{m,n}(g) \sum_{r \in \mathbb{Z}} c_H(r) e_q(r\overline{H}\rho) \widehat{F}\left(\frac{rJ}{Hq}\right) \right. \\ & \quad \left. - \sum_{r \in \mathbb{Z}} c_H(r) e_v(r\overline{H}\rho_2) \widehat{F}\left(\frac{rJ}{Hv}\right) \right] + O_{A,F,H,\eta_0}(X^{-A}). \end{aligned} \quad (69)$$

The joint sum is omitted when $\iota_{m,n}(g) = 0$. The error remains arbitrarily power-saving after all polynomially many row and divisor sums in the selected packet. The additive zero frequency in (69) is

$$J\widehat{F}(0) \frac{\varphi(H)}{H} \frac{g\iota_{m,n}(g) - 1}{uv}. \quad (70)$$

Proof. The four exact natural Poisson lattices are the joint, u -single, v -single, and constant lattices of TPC-20 [4]. In the normalization above their bracket is

$$\begin{aligned} & g\iota_{m,n}(g) \sum_r c_H(r) e_q(r\overline{H}\rho) \widehat{F}\left(\frac{rJ}{Hq}\right) \\ & - \sum_r c_H(r) e_u(r\overline{H}\rho_1) \widehat{F}\left(\frac{rJ}{Hu}\right) - \sum_r c_H(r) e_v(r\overline{H}\rho_2) \widehat{F}\left(\frac{rJ}{Hv}\right) + \sum_r c_H(r) \widehat{F}\left(\frac{rJ}{H}\right). \end{aligned}$$

By (62), every nonzero frequency in the u -single channel has Fourier argument at least X^{η_0} . The constant channel is shorter still. Repeated integration by parts removes their nonzero frequencies, including the fixed polynomial multiplicity. Their zero frequencies have coefficients -1 and $+1$, so they cancel. This proves (69). Since $c_H(0) = \varphi(H)$, the two remaining zero modes give (70). $\square$

Remark 3.2 (Why the long channel cannot be deleted). When v is comparable with or larger than J , the first nonzero Fourier argument $J/(Hv)$ need not grow. Smoothness alone therefore does not remove the second sum in (69). Bounding a complete long Möbius divisor shell pointwise gives only $X^{o(1)}$, not a fixed power of $\log X$; this distinction cannot be repaired by changing the letter ϵ .

3.2 Scalar terms restore the joint lattice

Define the two bare centered signals

$$Z_{m,S}(j) = \sum_{\substack{u \leq S \\ (u,mH)=1}} b_R(u) C_{m,u}(j), \quad (71)$$

$$V_{n,S,T}(j) = \sum_{\substack{S < v \leq T \\ (v,nH)=1}} a(v) C_{n,v}(j), \quad (72)$$

and the long scalar density

$$\alpha_{n,S,T} = \sum_{\substack{S < v \leq T \\ (v,nH)=1}} \frac{a(v)}{v} = A_{nH}(T) - A_{nH}(S). \quad (73)$$

The prefix identities of [proposition 2.2](#) say exactly

$$P_{m,S} = Z_{m,S} + E_S(m), \quad L_{n,S,T} = V_{n,S,T} + \alpha_{n,S,T}. \quad (74)$$

Hence

$$P_{m,S} L_{n,S,T} = Z_{m,S} V_{n,S,T} + E_S(m) V_{n,S,T} + \alpha_{n,S,T} Z_{m,S} + E_S(m) \alpha_{n,S,T}. \quad (75)$$

In the first term of (75), the long single channel of [theorem 3.1](#) is multiplied by

$$\sum_{\substack{u \leq S \\ (u,mH)=1}} \frac{b_R(u)}{u} = \delta_H(m) + E_S(m). \quad (76)$$

The remaining three scalar terms restore the channels removed by centering. After exact recombination, the coefficient $E_S(m)$ cancels rather than being bounded against a long divisor signal, and one recovers (63). Thus no estimate of the false form $X^\epsilon (\log X)^{-A} \ll (\log X)^{-B}$ is used.

Equivalently, one may apply Poisson summation directly to the raw joint indicator in (63). If the pair is compatible, the exact formula is

$$\begin{aligned} & \sum_{\substack{j \in \mathbb{Z} \\ (j,H)=1}} F(j/J) \mathbf{1}_{u|mj+h} \mathbf{1}_{v|nj+h} \\ &= \frac{J}{Hq} \sum_{r \in \mathbb{Z}} c_H(r) e_q(r \overline{H} \rho) \widehat{F} \left(\frac{rJ}{Hq} \right), \end{aligned} \quad (77)$$

and it is zero otherwise. The zero frequency of (77) is

$$J \widehat{F}(0) \frac{\varphi(H)}{H} \frac{g \iota_{m,n}(g)}{uv}. \quad (78)$$

This is the kernel treated in [section 4](#); every nonzero frequency is treated by the conductor rectangle in [section 5](#).

Remark 3.3 (The centered zero kernel). For comparison, the zero kernel of the bare product $Z_{m,S} V_{n,S,T}$ is obtained from (78) by replacing $g \iota_{m,n}(g)$ with $g \iota_{m,n}(g) - 1$. Its disconnected -1 part factors into the complete short prefix and $A_{nH}(T) - A_{nH}(S)$; the latter is an endpoint difference and has arbitrary logarithmic decay. The main proof uses the cleaner calibrated identity instead.

4 Shell-averaged control of the physical zero mode

The nonzero Ramanujan–Gauss estimate used later has no content at additive frequency zero. We therefore isolate that frequency before introducing multiplicative characters. For an off-diagonal row pair write $\Delta = m - n \neq 0$, and define

$$K_{S,T}^0(m,n) := \sum_{\substack{u \leq S, S < v \leq T \\ (u,mH)=(v,nH)=1}} \frac{b_R(u)a(v)}{uv} (u,v) \mathbf{1}_{(u,v)|\Delta}. \quad (79)$$

All effective summands are squarefree because both divisor coefficients are supported on squarefree integers.

Theorem 4.1 (Pointwise shell cancellation). *Suppose $S = X^{s+o(1)}$, $T \leq X^{C_T}$, where $s, C_T > 0$ are fixed. For every fixed $A > 0$ and $0 < \kappa < s$, uniformly over the retained off-diagonal row pairs,*

$$\boxed{K_{S,T}^0(m,n) \ll_{A,\kappa,C_T,h} (\log X)^{-A} + X^{-s+\kappa+o(1)}}. \quad (80)$$

The $o(1)$ in the second term comes only from the divisor bound for the nonzero row determinant.

Proof. Put $g = (u, v)$, $u = gc$, and $v = gd$. Exactness of the gcd and squarefree support make g, c, d pairwise coprime. Reordering the finite sum gives

$$\begin{aligned} K_{S,T}^0(m,n) &= \sum_{\substack{g|\Delta \\ g \text{ squarefree} \\ (g,mnH)=1}} \frac{1}{g} \sum_{\substack{c \leq S/g \\ (gc,mH)=1}} \frac{b_R(gc)}{c} \\ &\quad \times \sum_{\substack{S/g < d \leq T/g \\ (d,gcH)=1}} \frac{a(gd)}{d}. \end{aligned} \quad (81)$$

The modulus in the innermost coprimality condition is exactly gcH : the factors g, c come from $(u, v) = g$, and nH comes from target primitivity on the second row.

Split (81) at

$$G = SX^{-\kappa}. \quad (82)$$

If $g \leq G$, then $S/g \geq X^\kappa$. Fix g, c and set $M = gcH$. Since $gc \leq S$, $n \asymp Q$, and every scale is a fixed power of X , there is a fixed C_0 such that $M \leq X^{C_0}$. The coefficient identity

$$a(gd) = \mu(g)a(d) - \mu(g) \log g \mu(d) \quad (83)$$

therefore turns the inner sum into

$$\mu(g) \{A_M(T/g) - A_M(S/g)\} - \mu(g) \log g \{B_M(T/g) - B_M(S/g)\}. \quad (84)$$

By lemma 2.1, the main constants $M/\varphi(M)$ in the first difference cancel exactly; both differences have arbitrary logarithmic decay. Moreover,

$$\frac{M}{\varphi(M)} \ll \log \log(3X), \quad \sum_{g|\Delta} \frac{1}{g} \ll \log \log(3X),$$

and lemma 2.3 bounds the c -sum by a fixed power of $\log X$. Increasing the calibration exponent in terms of the desired A proves

$$K_{S,T}^{0,g \leq G}(m,n) \ll_A (\log X)^{-A}. \quad (85)$$

For $g > G$, use absolute values. The cofactor estimates

$$\sum_{c \leq S/g} \frac{|b_R(gc)|}{c} \ll (\log(2X))^4, \quad (86)$$

$$\sum_{d \leq T/g} \frac{|a(gd)|}{d} \ll (\log(2X))^2 \quad (87)$$

follow from [lemma 2.3](#) and a harmonic logarithmic sum. Since $\Delta \neq 0$,

$$\sum_{\substack{g|\Delta \\ g>G}} \frac{1}{g} \leq \frac{\tau(|\Delta|)}{G} = X^{-s+\kappa+o(1)}. \quad (88)$$

Combining (85)–(88) proves the theorem. $\square$

4.1 From one row pair to the packet

Let $\mathcal{Z}_{S,T}^0$ denote the contribution of (78) to the joint term in (63). Thus

$$\mathcal{Z}_{S,T}^0 = J\hat{F}(0) \frac{\varphi(H)}{H} \sum_{m,n} x_m y_n \mathbf{m}(m,n) K_{S,T}^0(m,n). \quad (89)$$

Corollary 4.2 (Mass-sensitive zero-mode theorem). *Under the hypotheses of [theorem 4.1](#),*

$$|\mathcal{Z}_{S,T}^0| \ll_{A,\kappa,F,H} J\mathcal{Z}_1 \left\{ (\log X)^{-A} + X^{-s+\kappa+o(1)} \right\}. \quad (90)$$

Consequently:

$$(i) \text{ under only (25),} \quad |\mathcal{Z}_{S,T}^0| \ll XQ \left\{ X^\epsilon (\log X)^{-A} + X^{-s+\kappa+\epsilon+o(1)} \right\}; \quad (91)$$

(ii) under (27), for every fixed $B > 0$,

$$|\mathcal{Z}_{S,T}^0| \ll_{B,K,F,H} XQ (\log X)^{-B}; \quad (92)$$

(iii) if $\hat{F}(0) = 0$, then $\mathcal{Z}_{S,T}^0 = 0$ identically, independently of the row mass.

Proof. Insert (80) into (89). Since $JQ^2 \asymp XQ$, the soft statement follows from (25). It is important that $X^\epsilon (\log X)^{-A}$ is retained: for fixed $A, \epsilon > 0$ it does not tend to zero. Under (27), choose the calibration exponent in [theorem 4.1](#) after B, K ; the large- g term has a fixed power saving because $\kappa < s$. The last statement is immediate from (89). $\square$

Remark 4.3 (Why the diagonal mask is structural). If $m = n$, then every g divides $\Delta = 0$, so (88) is unavailable. The row-diagonal removal is therefore not a cosmetic pruning inherited from an earlier argument; it is used directly in the zero-mode theorem.

5 The asymmetric exact-conductor rectangle

We now treat every nonzero additive frequency of the raw joint lattice (77). Only the upper bounds $u \leq S$ and $v \leq T$ enter the nonnegative energy estimates; the lower restriction $v > S$ may be discarded when enlarging such an estimate.

5.1 Pairwise character coordinates

For one squarefree pair use (66). Since $q = ud$ and $(u, d) = 1$, there is an exact product decomposition

$$\mathfrak{X}(q) \simeq \mathfrak{X}(u) \times \mathfrak{X}(d), \quad \chi \longmapsto (\alpha, \gamma), \quad (93)$$

and

$$\text{cond } \chi = (\text{cond } \alpha)(\text{cond } \gamma). \quad (94)$$

The shared factor g is assigned to the short-side coordinate $\alpha \bmod u$. Put

$$c_{m,n}^{(g)} = x_m y_n \mathbf{m}(m, n) \mathbf{1}_{m \equiv n \pmod{g}}. \quad (95)$$

For this fixed divisor pair, define its unweighted row fiber by

$$Z_{u,v}(\xi) = \sum_{\substack{m,n \\ g|m-n \\ \rho_{m,n}^{u,v} \equiv \xi \pmod{q}}} x_m y_n \mathbf{m}(m, n), \quad \xi \in \mathcal{U}_q, \quad (96)$$

and put

$$\bar{Z}_{u,v} = \frac{1}{\varphi(q)} \sum_{\xi \in \mathcal{U}_q} Z_{u,v}(\xi), \quad D_{u,v} = Z_{u,v} - \bar{Z}_{u,v}. \quad (97)$$

The exact pairwise transform from TPC-22 is

$$\hat{Z}_{u,v}^\times(\chi) = \overline{\chi(-h)} \sum_{m,n} c_{m,n}^{(g)} \alpha(m) \gamma(n). \quad (98)$$

There is no residual factor $1/\varphi(g)$, and the character on the second row is not conjugated [5].

Lemma 5.1 (Asymmetric compatibility energy). *Under (24) and (26),*

$$\boxed{\sum_{g \leq S} \sum_{m,n} |c_{m,n}^{(g)}|^2 \ll_\epsilon Q^2 X^\epsilon.} \quad (99)$$

No comparison between S and L is required.

Proof. The mask removes $m = n$. Hence

$$\sum_{g \leq S} \sum_{m,n} |c_{m,n}^{(g)}|^2 \leq \sum_{m \neq n} |x_m|^2 |y_n|^2 \sum_{g|m-n} 1.$$

The last multiplicity is at most $\tau(|m - n|) \ll_\epsilon X^\epsilon$, independently of S . Apply (24) and rename ϵ . $\square$

5.2 Tensor energy at the true conductors

We use the weighted exact-conductor large sieve in the following normalization. If $w \asymp W$ runs over squarefree ambient moduli, a_r is supported on an interval of length N , and the exact conductors lie in $f \asymp F$, then

$$\sum_{w \asymp W} \frac{1}{\varphi(w)} \sum_{\substack{\chi \in \mathfrak{X}(w) \\ \text{cond } \chi \asymp F}} \left| \sum_r a_r \chi(r) \right|^2 \ll_\epsilon X^\epsilon \left(F + \frac{N}{F} \right) \sum_r |a_r|^2. \quad (100)$$

TPC-22 derives this form from the primitive multiplicative large sieve at the true conductor [5]; the underlying large sieve is standard [1, Chapter 7]. Put

$$\mathcal{K}_Q(F) = F + \frac{Q}{F}. \quad (101)$$

For dyadic Y, F_1, F_2 , let

$$\begin{aligned} \mathcal{E}_{Y;S,T}(F_1, F_2) := & \sum_{\substack{g,c,d \text{ pairwise coprime} \\ gc \leq S, S < gd \leq T, gcd \asymp Y}} \frac{|b_R(gc)a(gd)|^2}{\varphi(gcd)} \\ & \times \sum_{\substack{\alpha \in \mathfrak{X}(gc) \\ \text{cond } \alpha \asymp F_1}} \sum_{\substack{\gamma \in \mathfrak{X}(d) \\ \text{cond } \gamma \asymp F_2}} \left| \sum_{m,n} c_{m,n}^{(g)} \alpha(m) \gamma(n) \right|^2. \end{aligned} \quad (102)$$

All inherited primitive restrictions are understood.

Theorem 5.2 (Asymmetric tensor conductor energy). *Uniformly for $Y \leq ST$, $F_1 \leq S$, and $F_2 \leq T$,*

$$\boxed{\mathcal{E}_{Y;S,T}(F_1, F_2) \ll_{\epsilon} Q^2 X^{\epsilon} \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}. \quad (103)$$

Proof. Fix g , split the two ambient moduli into dyadic ranges, and enlarge the nonnegative sum by dropping divisibility, coprimality, and shell restrictions after retaining those ranges. Apply (100) in the m -coordinate in its Hilbert-valued form, leaving the n -coordinate as a vector, and then apply it in the n -coordinate. Since both row variables lie in intervals of length $O(Q)$, the result for fixed g is

$$\ll_{\epsilon} X^{\epsilon} \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2) \sum_{m,n} |c_{m,n}^{(g)}|^2.$$

Sum in g , invoke lemma 5.1, and absorb the soft coefficient bounds and logarithmically many subdivisions into X^{ϵ} . $\square$

5.3 Two estimates for the same nonzero cell

For a function D on $\mathcal{U}_q$, use the multiplicative Fourier normalization

$$\widehat{D}^{\times}(\chi) = \sum_{\xi \in \mathcal{U}_q} D(\xi) \overline{\chi(\xi)}, \quad D(\xi) = \frac{1}{\varphi(q)} \sum_{\chi \in \mathfrak{X}(q)} \widehat{D}^{\times}(\chi) \chi(\xi).$$

For the fixed pair $u = gc, v = gd$, let $D_{u,v}^{F_1, F_2}$ be the inverse multiplicative transform of $\widehat{D}_{u,v}^{\times}$, restricted under (93), to

$$\text{cond } \alpha \asymp F_1, \quad \text{cond } \gamma \asymp F_2.$$

Define the corresponding retained-frequency scalar functional

$$\mathcal{L}_{u,v}^{F_1, F_2, \leq \omega} := \frac{J}{Hq} \sum_{0 < |r| \leq X^{\omega} Hq/J} c_H(r) \widehat{F} \left(\frac{rJ}{Hq} \right) \sum_{\xi \in \mathcal{U}_q} D_{u,v}^{F_1, F_2}(\xi) e_q(r \overline{H} \xi), \quad (104)$$

and the pair-separated cell aggregate

$$\mathfrak{L}_{Y;S,T}^{\leq \omega}(F_1, F_2) := \sum_{\substack{g,c,d \text{ pairwise coprime} \\ gc \leq S, S < gd \leq T, gcd \asymp Y}} |b_R(gc)a(gd)| |\mathcal{L}_{gc,gd}^{F_1, F_2, \leq \omega}|. \quad (105)$$

This is a sum of pairwise scalar functionals, not the norm of a fictitious global rectangular projection.

The restricted-frequency norm route, licensed by (58), gives

$$\mathfrak{L}_{Y;S,T}^{\leq \omega}(F_1, F_2) \ll_{\epsilon} Q X^{\epsilon} \sqrt{JY \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}. \quad (106)$$

There is a second estimate. Resolve both exact conductors before summing additive frequencies. If $f_1 = \text{cond } \alpha$, $f_2 = \text{cond } \gamma$, and $e = q/(f_1 f_2)$ is the inducing quotient, then at $r \neq 0$ the

induced Gauss sum is a primitive Gauss factor times the Ramanujan sum $c_e(r)$. The smooth nonzero-frequency estimate

$$\sum_{r \neq 0} |c_e(r)| \left| \widehat{F} \left(\frac{rJ}{Hq} \right) \right| \ll_{F,H} \frac{Hq}{J} \tau(e) \quad (107)$$

then gives

$$\mathfrak{L}_{Y;S,T}^{\leq \omega}(F_1, F_2) \ll_{\epsilon} Q F_1 F_2 X^{\epsilon} \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}. \quad (108)$$

This is the TPC-22 induced-Gauss argument with rectangular ambient ranges; restricting its left side to the retained frequencies only makes the absolute majorant smaller. Formula (107) must not include $r = 0$, because $c_e(0) = \varphi(e)$; that frequency was treated in [section 4](#).

Combining the two estimates for the same scalar cell gives

$$\boxed{\mathfrak{L}_{Y;S,T}^{\leq \omega}(F_1, F_2) \ll_{\epsilon} Q X^{\epsilon} \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)} \min\{\sqrt{JY}, F_1 F_2\}.} \quad (109)$$

One-sided principal coordinate cells $(1, F_2)$ and $(F_1, 1)$ remain genuine discrepancy cells.

Let $\mathscr{J}_{q;S,T}^{\text{disc}, \neq 0}$ denote the nonprincipal part of the joint $r \neq 0$ packet at denominator q . Linearity, exact conductor decomposition, and pairwise triangle inequality give the formal reassembly

$$\sum_{q \asymp Y} |\mathscr{J}_{q;S,T}^{\text{disc}, \neq 0}| \leq \sum_{\substack{F_1, F_2 \\ (F_1, F_2) \neq (1, 1)}} \mathfrak{L}_{Y;S,T}^{\leq \omega}(F_1, F_2) + O_A(X^{-A} \mathcal{Z}_1). \quad (110)$$

The last term is the single undecomposed smooth Fourier-tail error from (61). It is inserted before the exact-conductor projection; the retained cells above therefore never require a pair-separated tail estimate.

The global principal cell $(1, 1)$ is the Ramanujan mean. The rectangular form of the TPC-23 mean argument [6] gives

$$\sum_q |\mathscr{J}_{q;S,T}^{\text{mean}, \neq 0}| \ll_{\epsilon, F, H} \mathcal{Z}_1 X^{\epsilon}, \quad (111)$$

because

$$\sum_{u \leq S, v \leq T} \frac{1}{[u, v]} \ll \log^3(2ST).$$

Under (25), this is $Q^2 X^{\epsilon}$, a fixed power below $XQ = Q^2 J$ whenever J has fixed positive power.

Remark 5.3 (Long-side inequalities not used here). The conductor proof uses neither $T < J$ nor $L > 2T$. The shell lower bound $v > S$ only reduces the nonnegative energy. What remains essential is the divisor-independent off-diagonal row mask, the no-wrap interface for (106), and polynomial control of the total Fourier tail.

6 Exponent ledger and passage beyond the Poisson face

Write

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad S = X^{s+o(1)}, \quad T = X^{t+o(1)}, \quad (112)$$

with $0 < s \leq t$. On a denominator and conductor cell put

$$Y = X^{y+o(1)}, \quad F_i = X^{f_i+o(1)}, \quad k_i = \max\{f_i, \beta - f_i\}. \quad (113)$$

The rectangular support gives

$$0 \leq y \leq s + t, \quad 0 \leq f_1 \leq s, \quad 0 \leq f_2 \leq t, \quad (114)$$

and $\mathcal{K}_Q(F_i) = X^{k_i+o(1)}$.

The natural scale $XQ = Q^2J$ has exponent $1 + \beta$. The norm and direct-Gauss routes in (109) have exponents

$$\beta + \frac{1 - \beta + y + k_1 + k_2}{2}, \quad \beta + f_1 + f_2 + \frac{k_1 + k_2}{2},$$

respectively. Their saving exponents are therefore

$$\eta_A = \frac{\beta + 1 - y - k_1 - k_2}{2}, \quad \eta_B = 1 - f_1 - f_2 - \frac{k_1 + k_2}{2}. \quad (115)$$

Proposition 6.1 (Asymmetric three-cell conductor ledger). *Every cell in (114) satisfies*

$$\max\{\eta_A, \eta_B\} \geq \eta_{\text{asym}}(\beta; s, t), \quad (116)$$

where

$$\eta_{\text{asym}}(\beta; s, t) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 1 - 2s - 2t}{2}, \frac{3 - \beta - s - 5t}{4} \right\}. \quad (117)$$

Proof. Decreasing y improves η_A , so use $y = s + t$.

If $f_1, f_2 \leq \beta/2$, then $k_1 + k_2 = 2\beta - f_1 - f_2$, and

$$\eta_B = 1 - \beta - \frac{f_1 + f_2}{2} \geq 1 - \frac{3\beta}{2}.$$

If $f_1, f_2 \geq \beta/2$, then $k_1 + k_2 = f_1 + f_2 \leq s + t$, whence

$$\eta_A \geq \frac{\beta + 1 - 2s - 2t}{2}.$$

In a mixed cell write $x \leq \beta/2 \leq z$ and $d = z - x \geq 0$. Then $k_1 + k_2 = \beta + d$, so

$$\eta_A \geq \frac{1 - s - t - d}{2}. \quad (118)$$

Since $z \leq t$,

$$\eta_B = 1 - \frac{\beta}{2} - 2z + \frac{d}{2} \geq 1 - \frac{\beta}{2} - 2t + \frac{d}{2}. \quad (119)$$

The maximum is at least the average of the last two lower bounds, namely $(3 - \beta - s - 5t)/4$. Combining the three cases proves the claim. $\square$

6.1 The mass-sensitive mixed-increment theorem

Theorem 6.2 (One-sided calibrated-increment estimate). *Assume the row and target-primitivity interfaces of section 2, together with the divisor-independent off-diagonal mask, (58), and (61). The short-side condition (62) is not required for this calibrated formulation. Suppose $\eta_{\text{asym}}(\beta; s, t) > 0$ with a fixed margin and $0 < \kappa < s$. Then for every fixed $A > 0$ and every sufficiently small fixed $\epsilon > 0$, there is a fixed $\eta > 0$ such that*

$$|\mathcal{I}_{S,T}| \ll XQX^{-\eta} + J\mathcal{Z}_1 \left\{ (\log X)^{-A} + X^{-s+\kappa+o(1)} + L^{-1}X^\epsilon \right\}. \quad (120)$$

Here η may be any sufficiently small fixed number below the conductor margin and the principal-mean margin $1 - \beta$, after the auxiliary ϵ 's are chosen. The transpose packet obeys the same estimate.

Proof. Use [proposition 2.4](#). Reassemble the joint $r \neq 0$ discrepancy through (110). Apply [proposition 6.1](#) and (109); the fixed margin absorbs all dyadic and divisor multiplicities. The global principal cell is bounded by (111); under (25) it is $Q^2 X^\epsilon = XQX^{-(1-\beta)+\epsilon}$. The physical zero mode is (90), and the drift is (64). The inherited Fourier tail is made smaller than the displayed fixed-power term by choosing the smooth truncation order after its polynomial mass. This proves (120). $\square$

Corollary 6.3 (Two rigorous closure regimes). *Retain the hypotheses of [theorem 6.2](#), and suppose $L \geq X^{\lambda_0}$ for a fixed $\lambda_0 > 0$.*

(i) *If the total projective mass satisfies (27), then for every fixed $B > 0$,*

$$\boxed{\mathcal{I}_{S,T} \ll_{B,K,F,H} XQ(\log X)^{-B}}. \quad (121)$$

(ii) *Under only the soft mass hypothesis (25), if $\widehat{F}(0) = 0$, then some fixed $\eta' > 0$ satisfies*

$$\boxed{\mathcal{I}_{S,T} \ll XQX^{-\eta'}}. \quad (122)$$

after the conventional auxiliary X^ϵ losses are chosen smaller than the fixed margins.

Proof. In (i), choose the calibration exponent A after B, K . The large- g , drift, conductor, and principal-mean terms have fixed power savings. In (ii), the physical zero mode is identically zero by [corollary 4.2](#); every remaining term has a fixed-power margin. $\square$

Remark 6.4 (What the soft theorem does not say). With general $\widehat{F}(0) \neq 0$ and only $\mathcal{Z}_1 \ll Q^2 X^\epsilon$, the first term inside braces in (120) remains $XQX^\epsilon(\log X)^{-A}$. It is neither asserted to be $o(XQ)$ nor relabelled as logarithmic saving.

6.2 The general one-sided passage region

For a genuine crossing of the old Poisson face, require

$$s < 1 - \beta < t. \quad (123)$$

Positivity of the three entries in (117) gives the explicit window

$$\boxed{1 - \beta < t < \min \left\{ \frac{\beta + 1}{2} - s, \frac{3 - \beta - s}{5} \right\}}, \quad (124)$$

together with $\beta < 2/3$ and $s < 1 - \beta$, all with fixed margins. The interval in (124) is nonempty precisely when

$$\max \left\{ \frac{1 + 2s}{3}, \frac{2 + s}{4} \right\} < \beta < \min \left\{ \frac{2}{3}, 1 - s \right\}. \quad (125)$$

There exist β, t satisfying these inequalities if and only if

$$\boxed{s < \frac{2}{5}}. \quad (126)$$

These are sufficient faces for the certified three-cell ledger, not a claim that the two underlying cell bounds are globally optimal.

6.3 The baseline cutoff $S = R$

Take

$$s = \frac{1}{2} - \delta, \quad t = \frac{1}{2} - \delta + \xi. \quad (127)$$

Then (117) becomes

$$\eta_{\text{asym}}(\beta, \delta, \xi) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 4\delta - 1}{2} - \xi, \frac{6\delta - \beta - 5\xi}{4} \right\}. \quad (128)$$

At $\xi = 0$, this agrees with the TPC-22 square ledger; when $s = t$, the general formula agrees with the TPC-23 dynamic-square ledger.

The condition $T > J$ is

$$\xi > \frac{1}{2} + \delta - \beta. \quad (129)$$

Together with positivity of (128), this proves the passage window

$$\frac{1}{2} + \delta - \beta < \xi < \min \left\{ \frac{\beta + 4\delta - 1}{2}, \frac{6\delta - \beta}{5} \right\}, \quad (130)$$

under

$$\beta < \frac{2}{3}, \quad \beta < \frac{1}{2} + \delta. \quad (131)$$

The latter inequality keeps the baseline leg $S = R$ strictly below J . A simple nonempty subregion is

$$\delta > \frac{1}{10}, \quad \frac{5}{8} - \frac{\delta}{4} < \beta < \min \left\{ \frac{2}{3}, \frac{1}{2} + \delta \right\}. \quad (132)$$

For the exact rational sample

$$\delta = \frac{3}{20}, \quad \beta = \frac{31}{50}, \quad \xi = \frac{1}{25}, \quad (133)$$

we have

$$R = X^{7/20}, \quad J = X^{19/50}, \quad T = X^{39/100} > J,$$

and the three entries of (128) are

$$\frac{7}{100}, \quad \frac{7}{100}, \quad \frac{1}{50}.$$

Thus the certified joint nonzero saving is $1/50$.

7 Exact splicing, a failed shortcut, and scope

7.1 Where the mixed increment sits in the residual

Let U contain every positive target in the selected orbit packet. By (49),

$$P_{m,U}(j) = \Lambda(mj + h) - \Lambda_R(mj + h) - \delta_H(m). \quad (134)$$

For $R \leq S < T \leq U$, the prefix has the exact decomposition

$$P_{m,U} = P_{m,S} + L_{m;S,T} + L_{m;T,U}. \quad (135)$$

Therefore a two-row residual product expands into nine explicitly labelled pieces. The present theorem treats

$$P_{m,S}L_{n;S,T} \quad \text{and, by transposition,} \quad L_{m;S,T}P_{n,S}. \quad (136)$$

It does not estimate $L_{m;S,T}L_{n;S,T}$, either component involving $L_{\cdot;T,U}$, or the already studied base product $P_{m,S}P_{n,S}$. Consequently (121) is a closure of a genuine residual increment, but not a closure of the complete residual correlation from TPC-18 [2].

The shell lower endpoint is also structural. In (84), the main constant in $A_M(Y)$ disappears only because the long divisor variable is summed over the difference $(S/g, T/g]$. Replacing that shell by a single modulus destroys the mechanism.

7.2 Pointwise calibration at a fixed modulus is false

The last warning has a finite counterexample. For a cutoff R , define the centered response of the complete short prefix to one modulus v by

$$\Gamma_R(v; \Delta) = \sum_{u \leq R} \frac{b_R(u)}{u} \left((u, v) \mathbf{1}_{(u, v) | \Delta} - 1 \right), \quad (137)$$

temporarily suppressing row coprimality. Choose a prime $R/2 < p \leq R$, put $v = p$, and take $p \mid \Delta$. For every $u \leq R$ other than $u = p$, either $(u, p) = 1$, in which case the bracket vanishes, or $b_R(u) = 0$ for a nonsquarefree multiple of p . Since only $b = 1$ occurs in (32),

$$\lambda'_R(p) = -\frac{p}{p-1}, \quad b_R(p) = \log p + \frac{p}{p-1}. \quad (138)$$

Thus

$$\Gamma_R(p; \Delta) = \frac{p-1}{p} b_R(p) = \left(1 - \frac{1}{p}\right) \log p + 1 \asymp \log R. \quad (139)$$

Bertrand's postulate supplies such a prime for every sufficiently large R . Hence a uniform assertion $\Gamma_R(v; \Delta) = o(1)$ is false. This does not contradict theorem 4.1: that theorem retains the full outer Möbius shell and uses cancellation between its two endpoints.

7.3 What is unconditional, conditional, and still open

The result has three logically different levels.

- (1) The prefix, four-channel, recombination, and gcd identities are finite and exact. So are the fixed-modulus counterexample and the exponent ledger.
- (2) The nonzero conductor theorem and pointwise zero-shell theorem are proved without conjectural number-theoretic input for packets satisfying the displayed primitive, smoothness, no-wrap, and fixed-power-range interfaces. They use the primitive multiplicative large sieve and the classical zero-free region for $\zeta(s)$, not a Chowla or Elliott conjecture.
- (3) Full logarithmic closure for general $\hat{F}(0) \neq 0$ additionally uses the explicit fixed-projective-mass hypothesis (27). Under only the generic soft mass, the honest conclusion is (120); the mean-zero test-function corollary (122) remains fully closed.

In particular, nothing here proves a Hardy–Littlewood asymptotic, a positive lower bound for prime pairs, the twin-prime conjecture, or a general violation of the sieve parity barrier. The phrase “passage beyond the Poisson boundary” refers only to the support coordinate $T > J$ in the calibrated mixed increment.

7.4 The next precise gates

The theorem leaves several separable research problems.

- (i) Replace absolute use of $\mathcal{Z}_1$ in (90) by a row-averaged operator or Hilbert–Schmidt estimate for the small- g kernel. Such an estimate would remove the fixed-projective-mass hypothesis without requiring pointwise fixed-modulus calibration, which (139) rules out.
- (ii) Treat the both-new square $L_{m;S,T}L_{n;S,T}$. It has two long legs and no calibrated short prefix, so the present one-sided algebra does not apply.
- (iii) Reassemble the remaining $L_{\cdot;T,U}$ complement fibers. TPC-23 identifies their affine Möbius parity cocycle; polynomially uniform cancellation over those growing fibers remains open [6].
- (iv) Only after these analytic components have compatible bounds may one return to the global endpoint and positivity ledgers. A prime-pair main term is a separate stage, not a consequence of mixed shell cancellation alone.

Each gate has a valid negative fallback: failure of a proposed uniform estimate can be recorded by an exact obstruction such as (139), without being presented as evidence for or against the twin-prime conjecture.

8 Exact certificate and conclusion

The analytic arguments above are symbolic. We include a deterministic finite certificate to audit the identities and rational exponent ledger most vulnerable to sign, gcd, normalization, or boundary errors. It does not test the zero-free region, estimate a Möbius shell asymptotically, or sample primes for evidential purposes.

8.1 Certificate

The standard-library script

`experiments/tpc24_certificate.py`

uses `fractions.Fraction` throughout. It represents $\log n$ on the formal prime basis and represents products of two divisor coefficients as exact degree-two polynomials in those formal logarithms. It checks:

- (1) the exact finite-model density (36) on four primitive row models;
- (2) both calibrated-prefix identities (47) and (48) on the same models;
- (3) the direct physical zero kernel (79) against its exact gcd reparametrization (81) on four finite families;
- (4) the coefficient split (83) on every tested coprime squarefree pair;
- (5) the asymmetric three-cell inequality on an exact rational grid;
- (6) the specialization from $(\beta; s, t)$ to (β, δ, ξ) , including the rational crossing sample (133);
- (7) the hard-failure response (139), which is required to be nonzero with the predicted formal $\log p$ -coefficient.

Independent ordinary and `python -O` runs perform 55,084 exact checks and produce byte-identical JSON. The hashes are

JSON SHA-256	f638ca06fa641a570277b89b1ed17d81aeac2eaa264cf80d5a317378efbd0fc6
source SHA-256	7b9ce64a93c5abb24c02ca9da6ea763fa5fbd97192bc1fd700fe47a7887ad33d
certificate digest	c684a7c10e7cf3bd2cc46f2305af94d8452f018494d445ecdd3483325df6f50d

The JSON records explicitly that the projective-mass condition is a hypothesis and that the mass-sensitive zero term has not been deleted. Its flags for asymptotic evidence, full residual closure, Hardy–Littlewood, twin primes, and a general parity breakthrough are all false.

8.2 Conclusion

The old Poisson-short boundary is not a wall for every residual component. Once the object is organized as an exact calibrated-prefix increment, the misleading long single channel cancels algebraically. The remaining nonzero joint spectrum has a rectangular true-conductor decomposition, whose certified saving is (117). Its physical zero mode has a different mechanism: the complete long Möbius shell cancels the common endpoint constant for small shared factor, while the off-diagonal row determinant controls large shared factor.

This proves a genuine support advance. The explicit region (130) contains parameters with $S = R < J < T$, for example (133). The advance is stronger than a numerical observation: the nonzero spectrum has fixed-power saving, and the zero kernel has the pointwise theorem (80).

The paper also preserves the exact remaining boundary. Pointwise shell cancellation becomes a packet estimate with the actual absolute row mass $\mathcal{Z}_1$. Fixed-polylogarithmic projective mass yields the full logarithmic closure (121); a mean-zero test function yields the soft-mass closure (122). Generic soft mass and nonzero physical mean leave the explicit term in (120). Calling that term negligible would require a new row-averaged zero-mode estimate, not a change of notation.

The next useful target is therefore precise: prove cancellation of the small- g zero kernel against the actual separated row family, or derive a fixed-polylog projective decomposition for the relevant packet, and then attack the both-new square. Neither step is supplied here. In particular, this one-sided passage is not a twin-prime theorem and is not described as a general breach of parity.

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